

UCT302: COMPUTER ORGANIZATION & ARCHITECTURE

L	T	P	Cr
3	0	2	4.0

Course Objectives: To provide an understanding of the basic concepts of computer architecture and organization of the basic computer modules viz controls unit, central processing unit, input-output organization and memory unit.

Revision of basics in Boolean logic and Combinational/Sequential Circuits.

Functional blocks of a computer: CPU, memory, input-output subsystems, control unit.

Instruction set architecture of a CPU: Registers, instruction execution cycle, RTL interpretation of instructions, addressing modes, instruction set. Outlining instruction sets of some common CPUs.

Data representation: Signed number representation, fixed and floating-point representations, character representation.

Computer arithmetic: Integer addition and subtraction, ripple carry adder, carry look-ahead adder, etc. multiplication – shift-and-add, Booth multiplier, carry save multiplier, etc. Division restoring and non-restoring techniques, floating point arithmetic, IEEE 754 format.

Introduction to x86 architecture.

CPU control unit design: Hardwired and micro-programmed design approaches, design of a simple hypothetical CPU.

Memory system design: Semiconductor memory technologies, memory organization.

Peripheral devices and their characteristics: Input-output subsystems, I/O device interface, I/O transfers – program controlled, interrupt driven and DMA, privileged and non-privileged instructions, software interrupts and exceptions. Programs and processes – role of interrupts in process state transitions, I/O device interfaces – SCII, USB

Pipelining: Basic concepts of pipelining, throughput and speedup, pipeline hazards.

Parallel Processors: Introduction to parallel processors, Concurrent access to memory and cache coherency.

Memory organization: Memory interleaving, concept of hierarchical memory organization, cache memory, cache size vs. block size, mapping functions, replacement algorithms, write policies.

Laboratory work: Installing software development toolkit for ARM processor-based microcontrollers, Assembly language programming for ARM processors.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Illustrate various elementary concepts of computer architecture including syntax of register transfer language, micro operations, instruction cycle, and control unit.
2. Comprehend the design of basic computer using instruction formats & addressing modes.
3. Identify various memory management techniques and algorithms for performing addition, subtraction and multiplication etc.
4. Acquire the knowledge about pipelining, multiprocessors, and input-output organization.

Text Books:

1. M. M. Mano, Computer System Architecture, Prentice Hall of India, New Delhi, (1993) 3rd ed..
2. David A. Patterson and John L. Hennessy. Computer Organization and Design: The Hardware/Software Interface (19993), 2nd ed..
3. Carl Hamacher. Computer Organization and Embedded Systems, (1998), 6th ed.

Reference Books:

1. John P. Hayes. Computer Architecture and Organization, (1998), 3rd ed.
2. William Stallings. Computer Organization and Architecture: Designing for Performance, (2002), 6th ed.
3. Vincent P. Heuring and Harry F. Jordan Computer System Design and Architecture, (1996), 2nd ed.